

Why is a document relevant? Understanding the relevance scores in cross-lingual document retrieval

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ABSTRACT

Modern cross-lingual document retrieval models are capable of finding documents relevant to the query. However, they do not have the capabilities for explaining why the document is relevant. This paper proposes a novel learning-to-rank model named LM-EMD that uses the multilingual BERT language model and Earth Mover's Distance (EMD) to measure the document's relevancy to the input query and provide interpretable insights into why a document is relevant. The model uses the query and document token's contextual embeddings generated with multilingual BERT to measure their distances in the embedding space, which are then used by EMD to calculate the document's relevance score and identify which document tokens contribute the most to its relevancy. We evaluate the model on five language pairs of varying degrees of similarity and analyze its performance. We find that the model (1) performs similar as the best performing comparing model on high-resource languages, (2) is less effective on low-resource languages, and (3) provides insight into why a document is relevant to the query.

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1. Introduction

The number of multilingual documents is increasing rapidly. Due to global connectivity, we are communicating and collaborating with people from all around the globe. Because of that, we are required to prepare documents and resources in the language the target audience will be able to understand; when the audience are locals, one could prepare the resource in their local language. But when presenting to a more general audience, for instance at a global conference, the resource is usually prepared in a more spoken language. Thus, the topic is presented in different languages. This especially applies to education and law; In education, a topic is described, adapted, translated, and published in various languages, which can be seen in the massive amounts of courses and textbooks that are available. In law, legislation and treaties are translated in the languages of countries the documents are relevant to, making them accessible to their lawyers and decision makers. Because of the huge amount of multilingual documents, one requires an effective solution to search through them.

Modern cross-lingual document retrieval models provide limited explanation about the relevancy of the document to the

query. With the introduction of the transformer based multilingual language models, cross-lingual information retrieval (CLIR) has gotten more interest in the research communities. More specifically, cross-lingual document retrieval focus on finding documents relevant to the query, where the document and query can be in different languages. Because of the language difference, the models must resolve to more advanced approaches than simple keyword matching for measuring document's relevance. These usually involve extracting textual features and representations that are language agnostic. The features are then used to output a single value which signals how relevant a document is to a query. However, this value does not explain why is the document relevant.

Understanding why a document is relevant is helpful for both during the development of document retrieval models and their usage. When developing the model, one would normally test its performance. For examples where the model does not return appropriate results, knowing why a document is assigned the given relevancy value is useful for understanding why the model does not work, and thus speeding the development process. Once the model is ready, it can be made available for its users. Sometimes, the users do not necessary know how to structure the query to find documents they are interested in. Seeing why a document is relevant gives insights into how to change the query to get the appropriate results. Thus, reducing the time they spend searching.

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1.1. Research objectives and contributions

The main aim of this work is to develop a cross-lingual document retrieval model that is able to both calculate the document's relevancy to the query with high precision, and provide insight into why a document is relevant.

The main research objectives are to (1) evaluate the model's performance on data sets consisting of queries and documents in different languages (language pair data sets), where the languages have various degrees of similarity, (2) assess the degree of document score interpretability the model provides, and (3) analyze the impact the objective loss function used during training has on the performance of cross-lingual document retrieval models.

We propose a learning-to-rank model named LM-EMD which uses multilingual BERT to generate token vector representations and Earth Mover's Distance (EMD) to measure the document's relevancy to the query, and provide interpretable view into why a document is relevant.

The model is evaluated on five language pairs, three of which are high-resource, and two are low-resource languages, using the Precision at 1 (P@1) and Mean Average Precision (MAP) metrics. The results show the proposed model performs similar as the selected comparing models on high-resource languages, but its performance decreases on low-resources languages.

We also perform an analysis on how the selection of the objective loss function used during training effects the performance of the document retrieval models. We compare two commonly used loss functions, cross-entropy and pairwise ranking, and find that training the models using pairwise ranking yields better performance.

The contributions of this paper are:

- the theoretical background of using optimal transport for measuring document relevancy,
- a supervised multilingual learning-to-rank model which outputs interpretable relevance scores,
- a comparison of the model's performance fine-tuned using two loss functions, cross-entropy and pairwise ranking, commonly used for training document retrieval models that employ language models.

1.2. Paper structure

The paper is structured as follows: Section 2 contains an overview of the related work on cross-lingual information retrieval and EMD. Next, the background on the theory of optimal transport and how it is used to measure the distance between a document and query is in Section 3, followed by the proposed model's architecture description in Section 4. We present the experiment setting in Section 5. The experiment results are found in Section 6, followed by its discussion in Section 7. Finally, we conclude the paper and provide ideas for future work in Section 8.

2. Related work

General information retrieval. The objective of an efficient Information Retrieval (IR) method is to retrieve documents that are relevant to the user's query. To improve its performance, various methods are employed. Relevance feedback is a feature of an IR system which gathers user's feedback to improve the system's performance. With the emergence of neural information retrieval models, neural relevance feedback approaches were also developed [1]. Query expansion is a technique which extends the query by adding values to improve document retrieval. The added values are usually similar or related to those in the query, and can be assigned from knowledge bases [2,3], ontologies [4],

or by using word embeddings [5]. An extensive survey of query expansion techniques is found in [6]. In addition to developing an effective model for finding relevant documents, research on other aspects, such as fairness of the retrieved documents [7], and the experiment reproducibility on neural information retrieval models [8], has been done in the last years.

Prior work is different from the proposed model in the following ways: (1) our model does not use relevance feedback or query expansion, but uses the raw query text when calculating the document's relevance; (2) while prior work is evaluated on monolingual data sets, our method is designed for cross-lingual document retrieval, i.e. where the query and document are in different languages; (3) our method does not use knowledge bases or ontologies to retrieve similar or related tokens, but a multilingual language model that returns contextual embeddings of the query's and document's tokens. These are then used to measure their semantic similarity.

Cross-lingual information retrieval (CLIR). In order to retrieve documents that are in various languages, one must bridge the lexical gap between them. Because of this gap, traditional information retrieval models need to be accompanied with methods that are able to transfer the documents and queries into the same or similar semantic space. The most common are: machine translation systems, word embeddings, and transformer language models.

Machine Translation systems (MT) are used to translate either the query or documents [9]. They require huge amounts of parallel data to train the models where such resources are scarce for many language pairs. If needed, it is possible to employ MT services from different providers, such as Google Translate,¹ DeepL,² and Translate.com,³ but can be expensive when the data set is huge.

Word embeddings are used to generate query and document vector representations in the same semantic space. In recent years, various word embedding methods were proposed [10–13]. While most of them are trained on single language data sets, unsupervised approaches for aligning embeddings in a common vector space were explored [14–20]. The aligned word embeddings are then used to either (1) translate the query into the target languages and use traditional IR methods to find relevant documents, or (2) create a vector representation of both the query and document and calculate the cosine distance of the two vectors [21]. However, word embeddings assign the same vector to a word regardless of its meaning. This can make documents seem close in the semantic space, even though the meaning of their shared words are different.

Language models are used to generate query and document vector representations. The majority of research is on modifying BERT [22] for monolingual information retrieval [23–29]. These models measure the document's relevance by concatenating the query and document text, i.e. constructing the input in the form of [CLS] query [SEP] document [SEP], sending it through the model, and using the representation of the [CLS] token to predict the document's relevance. This approach enables sharing of the query and document contexts with each other, but is also expensive to compute since it has to be done for every query and document. An alternative approach was presented [30] where the query and document representations are constructed separately by pooling or aggregating the language model's output. One can then create the document representations ahead of time and generate only the query vectors at run-time. Multilingual language

¹ <https://translate.google.com/>.

² <https://www.deepl.com/en/translator>.

³ <https://www.translate.com/>.

models [31–33] have also been used for CLIR. For instance, multilingual BERT was used to generate inputs of the classification layer for predicting if a document is relevant to the query [34].

Our model uses a multilingual language model to extract the query and document token's contextual embeddings used to measure their semantic similarity. The model extracts the query and document token embeddings separately to reduce the computational cost of the methodology, similar as in [30].

Earth Mover's Distance (EMD). EMD, also known as the 1-Wasserstein distance [35,36], is a metric that measures the similarity between two distributions. It was widely adopted in image processing and retrieval [37–39]. However, it also has applications in natural language processing and text retrieval tasks.

The EMD was integrated in the pooling layers to reduce the error of an automatic short answer grading method [40]. In model distillation, EMD is used to calculate the minimum cumulative cost to transform the knowledge from the teacher to the student neural network [41]. Most relevant to our work, a modified EMD is used to measure the similarity of two sentences based on their associated word embeddings [42]. The evaluation was performed on monolingual data sets and showed that their approach outperformed the comparing models.

To the best of our knowledge, our work is the first that uses EMD in combination with a multilingual language model to measure document relevance scores on cross-lingual data sets. In addition, we leverage EMD for interpreting the relevance score and identifying tokens that contribute to the document's relevancy the most.

3. Measuring document relevancy with optimal transport

In this section, we present the theory used to measure the document's relevance to the query. First, we provide an overview of the optimal transport theory. Next, we describe how the queries and documents are represented, and how optimal transport is used to measure the document's relevance.

3.1. Background on optimal transport

Optimal transport (OT) is a powerful approach for comparing probability distributions [35,36]. Suppose we have two Polish spaces⁴ X and Y . Given the probability measures α on X and β on Y , we define the *coupling* of (α, β) as a probability measure $\pi \in X \times Y$ such that $(\text{pr}_X)_\# \pi = \alpha$ and $(\text{pr}_Y)_\# \pi = \beta$. The *optimal transport problem* asks for an optimal coupling that solves the minimization problem

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y), \quad (1)$$

where $\Pi(\alpha, \beta) := \{\pi \in X \times Y : (\text{pr}_X)_\# \pi = \alpha, (\text{pr}_Y)_\# \pi = \beta\}$ is the set of all admissible couplings, and $c : X \times Y \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$ is a cost function that is at least semi-continuous.

When the probability measures are defined on the same space, the optimal transport problem induces a notion of a distance on the space of probability measures. Suppose X is a Polish space, and α, β are probability measures on X . Given the ground metric $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ and the cost function $c(x, y) = d(x, y)^p$, the *p-Wasserstein distance* is defined as

$$W_p(\alpha, \beta) := \mathcal{L}_p(\alpha, \beta)^{1/p}. \quad (2)$$

It can be shown that W_p truly defines a metric on the space $\mathcal{P}(X)$ of all probability measures on X [35]. In addition, the optimal coupling π that solves $W_p(\alpha, \beta)$ gives us insight into how to

redistribute the mass between α and β , which is helpful for understanding the practical meaning of the solution. Thus, the *p*-Wasserstein distance is a powerful tool for measuring distances between probability measures.

From the *Kantorovich duality* [43] it follows that the optimal transport problem admits a dual problem which is more tractable in terms of computational complexity. Specifically, in the case of discrete measures the dual problem can be solved with linear programming. Let us state one possible version of the Kantorovich duality.

Let (X, α) and (Y, β) be probability spaces, and $c : X \times Y \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$ be a continuous cost function, such that

$$c(x, y) \geq \phi(x) + \psi(y), \quad \forall (x, y) \in X \times Y, \quad (3)$$

for some continuous functions $\phi : X \rightarrow \mathbb{R}$ and $\psi : Y \rightarrow \mathbb{R}$. Let us define the set of all pairs of functions (ϕ, ψ) satisfying Eq. (3) as

$$\mathcal{A}_{X \times Y}(c) = \{(\phi, \psi) : c(x, y) \geq \phi(x) + \psi(y), \forall (x, y) \in X \times Y\}.$$

Then, the Kantorovich duality is shown as

$$\mathcal{L}_c(\alpha, \beta) = \sup_{(\phi, \psi) \in \mathcal{A}_{X \times Y}(c)} \left(\int_Y \psi(y) d\beta(y) - \int_X \phi(x) d\alpha(x) \right). \quad (4)$$

Thus, the solution of the dual problem induces a solution of the original optimal transport problem.

In practice, when we compute the optimal transport problem we often add an entropic regularization factor to the right-hand side of Eq. (4) to make it easier to compute. However, it yields an approximate solution of the original problem. In the discrete and discretized continuous case, the regularized OT can be solved using the Sinkhorn algorithm [44] which is an iterative matrix-scaling procedure.

Probabilistic view of optimal transport. The optimal transport problem can be defined using probabilistic terms. Given two probability spaces (X, α) and (Y, β) , we define a *coupling* of (α, β) as a construction of two random variables (A, B) such that $\text{law}(A) = \alpha$ and $\text{law}(B) = \beta$, where for an arbitrary random variable X we can think of $\text{law}(X)$ as the probability distribution of X . Note that a coupling of (α, β) always exists [35].

The probabilistic equivalent to Eq. (1) is solving

$$\mathcal{L}_c(\alpha, \beta) = \inf_{(A, B) \in \Pi_p(\alpha, \beta)} \mathbb{E}(c(A, B)),$$

where $\Pi_p(\alpha, \beta) = \{(A, B) : \text{law}(A) = \alpha, \text{law}(B) = \beta\}$ is the set of all admissible couplings defined in probabilistic terms, and $c : X \times Y \rightarrow \mathbb{R}$ is the cost function. Similarly, the *p*-Wasserstein distance has an equivalent probabilistic formulation written as

$$W_p(\alpha, \beta) := \inf_{(A, B) \in \Pi_p(\alpha, \beta)} \mathbb{E}(c(A, B)^p)^{1/p}. \quad (5)$$

Using this definition of *p*-Wasserstein distance we can easily see that it indeed defines a metric on the space $\mathcal{P}(X)$ of probability measures on X [35]. In order to use *p*-Wasserstein distance for measuring relevancy between random variables, we must first introduce the *Prokhorov distance*. One possible definition is

$$d_p(\alpha, \beta) = \inf \left\{ \epsilon > 0 : \inf_{(A, B) \in \Pi_p(\alpha, \beta)} \mathbb{P}[d(A, B) > \epsilon] \leq \epsilon \right\}.$$

The distance $d_p(\alpha, \beta)$ is precisely the minimum distance “in probability” between random variables distributed according to the probability measures α and β [45]. The relation between the *p*-Wasserstein and Prokhorov distances is described as

$$(d_p)^2 \leq W_p \leq (1 + \text{diam}(X)) d_p, \quad (6)$$

where $\text{diam}(X) = \sup \{d(x, y) : x, y \in X\}$ [46]. This relation gives the means for interpreting the relevancy of two random variables:

⁴ A Polish space is a separable completely metrizable topological space.

Suppose we are given a metric space X , such that the diameter $\text{diam}(X)$ is sufficiently small, and the space $\mathcal{P}(X)$ of probability measures on X . Let α, β be probability measures on X , and A, B be random variables such that $\text{law}(A) = \alpha$ and $\text{law}(B) = \beta$. Then a smaller p -Wasserstein distance between α and β implies that the probability that B is lying in the complement of the ϵ -ball centered at A , $\mathbb{B}(A, \epsilon) := \{C : d(A, C) < \epsilon\}$, is sufficiently small. Thus, we can conclude that there must be some relation between the random variables A and B .

3.2. Measuring the document relevance to a query

Lets describe how documents and queries are represented. Let $G = (V, E)$ be a fully connected graph where the vertices $V = \{t_i\}_{i=1}^N$ are document tokens, and the edges $E = \{(t_i, t_j, w_{ij})\}_{i,j=1}^{N,N}$ are weighted with the distance between the edge's endpoints $w_{ij} = d(t_i, t_j)$. When the tokens are represented by vectors, the distance between them can be measured with the *cosine distance* defined as

$$d(t_i, t_j) = 1 - \frac{\langle \vec{v}_{t_i}, \vec{v}_{t_j} \rangle}{\|\vec{v}_{t_i}\|_2 \|\vec{v}_{t_j}\|_2} \in [0, 2],$$

where $\vec{v}_{t_i}, \vec{v}_{t_j}$ are the vector representations of the tokens t_i, t_j , respectively. Thus, the diameter of the graph G satisfies the following upper bound:

$$\text{diam}(G) = \sup_{i,j} \{d(t_i, t_j)\} \leq 2.$$

In order to define the probability measures on the graph G , we will use the Dirac delta function $\delta : G \rightarrow \{0, 1\}$, which we define as

$$\delta_x(y) = \begin{cases} 1, & x = y, \\ 0, & x \neq y, \end{cases} \quad (7)$$

where $x, y \in V$ are vertices in the graph G . The query q and document d can each be represented with a probability measure on G defined as

$$\mu_q = \sum_{i=1}^n w_{q_i} \delta_{q_i}, \quad \nu_d = \sum_{j=1}^m w_{d_j} \delta_{d_j},$$

where $\delta_{q_i}, \delta_{d_j}$ are the Dirac delta functions, and w_{q_i}, w_{d_j} are the weights associated with the tokens q_i, d_j , respectively. Since the diameter $\text{diam}(G)$ is small enough, and we can always find a coupling (Q, D) of (μ_q, ν_d) , we may use the criterion given by Eq. (6). Thus, if the p -Wasserstein distance $W_p(\mu_q, \nu_d)$ is (sufficiently) small, the probability $\mathbb{P}(D \in \mathbb{B}(Q, \epsilon))$ is high; meaning the random variable D , associated with document d , has a high probability of lying in the ϵ -ball centered at the random variable Q , associated with query q . Hence, we can conclude that a small $W_p(\mu_q, \nu_d)$ value implies that document d is relevant to query q . In addition, because μ_q and ν_d are discrete measures, we can calculate the p -Wasserstein distance by using the optimal transport's dual problem formulation, which can be solved with linear programming.

4. Model architecture

Using the theory presented in Section 3, we propose a learning-to-rank model that generates interpretable document relevance scores. It consists of two components: (1) a multilingual language model used to extract the query and document token representations, and (2) the EMD to calculate the document relevance scores. As shown in Section 4.3, the EMD solution can also be used to interpret why the document is relevant.

The model's architecture and data flow is depicted in Fig. 1. While this section focuses on the model's architecture, its technical implementation is found in Section 5.4.

4.1. Query and document representation

Language models generate contextual embeddings, i.e. representations that change based on the token's context. The language model first splits the input text into subword segments (known as tokens) by using the model's associated tokenizer, a subword segmentation algorithm that converts the input text into character n-grams. Afterwards, the model generates a vector representation for each token (known as an embedding) based on the token's context. These embeddings capture the token's semantic meaning. Therefore, tokens with similar or related meaning are found in similar contexts, hence their corresponding embeddings are closer in the embedding space. This can be extended to other languages by using multilingual language models, such as multilingual BERT [31], XLM [32], and XLM-RoBERTa [33].

Let q and d be the query and document, respectively. Using a language model we extract the query's and document's tokens and their embeddings. Because the model has a length limit for its input, we split longer documents into parts such that each part has at most the maximum length the model can process, and afterwards concatenate the model's outputs.

The tokens and their embeddings are then grouped together to create the query and document representation sets T_q, T_d consisting of (token, embedding) pairs used for measuring the document's relevance score (see Section 4.2):

$$T_q = \{(q_i, \vec{v}_{q_i})\}_{i=1}^n, \quad T_d = \{(d_j, \vec{v}_{d_j})\}_{j=1}^m,$$

where n, m are the number of extracted tokens, q_i, d_j are the tokens extracted from the query q and document d , and $\vec{v}_{q_i}, \vec{v}_{d_j}$ are the contextual embeddings of their corresponding tokens, respectively. Although the query and document can share the same tokens, their associated embeddings can be different since they reflect the token's meaning.

4.2. Measuring the relevance score

EMD, also known as the 1-Wasserstein distance, is used to measure the distance between two probability measures. We define the document relevance score as the EMD between the query and the document, such that smaller EMD imply greater relevance. Let $D \in \mathbb{R}^{n \times m}$ be the matrix of distances between the query and document tokens. Similar to previous work [42], we use token embeddings and calculate their cosine distances:

$$D_{i,j} = d(q_i, d_j) = 1 - \frac{\langle \vec{v}_{q_i}, \vec{v}_{d_j} \rangle}{\|\vec{v}_{q_i}\|_2 \|\vec{v}_{d_j}\|_2} \geq 0. \quad (8)$$

Let μ_q and ν_d be the query's and document's respective token distributions:

$$\begin{aligned} \mu_q &= \sum_{i=1}^n w_{q_i} \delta_{q_i}, & \sum_{i=1}^n w_{q_i} &= 1, \\ \nu_d &= \sum_{j=1}^m w_{d_j} \delta_{d_j}, & \sum_{j=1}^m w_{d_j} &= 1, \end{aligned} \quad (9)$$

where $\delta_{q_i}, \delta_{d_j}$ are the Dirac delta functions as defined in Eq. (7), and the weights w_{q_i}, w_{d_j} represent the mass of their corresponding tokens q_i, d_j , respectively.

The weights can reflect different aspects of their tokens. For instance, they can be the TF-IDF scores of the tokens within a given data set. However, because TF-IDF scores are dependent on the data set the model is applied on, it cannot be measured for out-of-vocabulary tokens, i.e. tokens that are not present in the training set. In addition, under-represented tokens, those that are rare in the training set, might get a TF-IDF score that wrongly

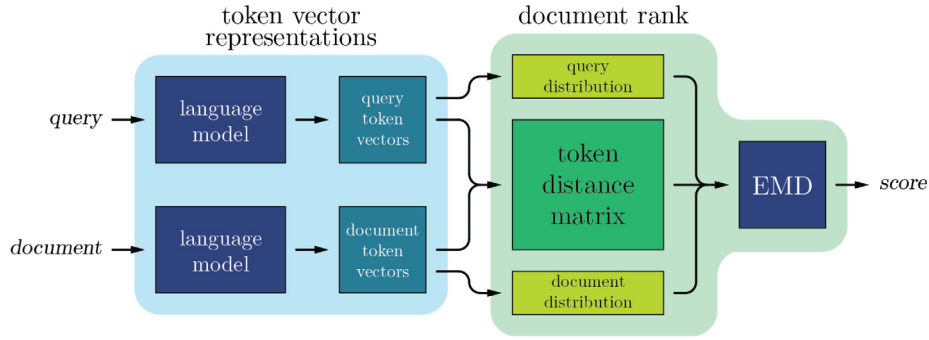


Fig. 1. The LM-EMD model architecture. The first step consists of retrieving token representations, which are used in the second step to calculate the Earth Mover's Distance between the query and document.

reflects the token's importance due to its lack of presence in the data set.

To facilitate handling both out-of-vocabulary and under-represented tokens, we set the weights to be uniformly distributed, i.e. $w_{q_i} = \frac{1}{n}$, $i = 1, \dots, n$ and $w_{d_j} = \frac{1}{m}$, $j = 1, \dots, m$. The uniform distribution will give equal importance to their respective query and document tokens, including to those that are out-of-vocabulary and under-represented, which allows the algorithm to take into consideration all of the provided tokens. Note that the algorithm permits non-uniform weight distribution as long as the weight constraints stated in Eq. (9) are met.

Using the distance matrix D and token distributions μ_q, ν_d , we calculate the EMD between the query and document by finding the matrix $P \in \mathbb{R}_{\geq 0}^{n \times m}$ that solves the following optimization problem:

$$\begin{aligned} \text{EMD}(q, d) &= \min_P \sum_{i=1}^n \sum_{j=1}^m P_{i,j} D_{i,j}, \\ \text{subject to: } &\sum_{j=1}^m P_{i,j} \leq w_{q_i}, \forall i \in \{1, \dots, n\}, \\ &\sum_{i=1}^n P_{i,j} \leq w_{d_j}, \forall j \in \{1, \dots, m\}, \\ &\sum_{i=1}^n \sum_{j=1}^m P_{i,j} = 1 = \min \left\{ \sum_{i=1}^n w_{q_i}, \sum_{j=1}^m w_{d_j} \right\}, \end{aligned} \quad (10)$$

Note that Eq. (10) is the optimal transport problem formulation in the discrete case for the cost matrix $C \in \mathbb{R}^{n \times m}$ with the entries $C_{i,j} = D_{i,j}$ given by Eq. (8). The solution P is called the *transportation matrix*, and shows the optimal way of “transporting” the mass from the distribution associated to the query to the distribution associated to the document. Because the distributions correspond to tokens, the transportation matrix can also be viewed as the optimal way of matching query's tokens with document's tokens w.r.t. the token distance matrix D (more is described in Section 4.3).

After calculating all of the document relevance scores, the model assigns the document ranks; documents that have lower EMD score are assigned a higher rank. The model then sorts the documents based on their rank and outputs the final order. Fig. 2 shows the overall document ranking steps.

4.3. Interpreting document relevance

In addition to being used to calculate the document's relevance score, the distance matrix and transportation matrix provide insight into why was such a score assigned to the document.

The solution of the optimal transport problem given by Eq. (10) can be analyzed by looking at the entries of the transportation matrix P . Because of the problem's constraints, the matrix P is non-zero and has at least one positive entry. The entries of P show how much “mass” is transported between the query and document token distributions. For instance, the entry $P_{i,j}$ shows the amount of “mass” that is moved between query token q_i and document token d_j ; the smaller the entry's value, the smaller the “mass” moved between the tokens. In order to reach the minimum of the optimal transport problem, the matrix P must have larger values at entries associated with the tokens that are closer in the embedding space (the entry $P_{i,j}$ must be large when the distance $D_{i,j}$ is small), otherwise it would transport “mass” between tokens that are further apart (have a higher move cost) and thus the matrix could not be the solution of the optimization problem. Furthermore, the larger the number of query and document tokens that are close in the embedding space, the smaller the EMD score – making the document more relevant to the query.

To view which document tokens have the largest significance on the relevance score, we must only read the values of the transportation matrix. Fig. 3 shows an illustrative example of how to interpret the token importance looking at the distance matrix and transportation matrix. The distance matrix contains various degrees of similarity between the tokens. The majority are distant but there are a few that are close, e.g. the token pair (q_4, d_5) exhibit strong similarity. This is then reflected in the transportation matrix which indicates a strong match between the pair, hence the token d_5 is significant for the document's relevance. Similarly, we find that (q_3, d_4) also have a strong match. While the (q_1, d_7) and (q_2, d_8) pairs have a medium match, the remainder have only a weak match. Hence, we conclude that the document is relevant because its tokens d_4, d_5, d_7 and d_8 have a significant similarity to the query tokens.

5. Experiments

We now present the experiment's setting. We introduce the data sets and how they are prepared for the experiments. Next, we describe the implementation details. Finally, we present the comparing methods used in the evaluation.

5.1. Data set

While the monolingual document retrieval task has a variety of open data sets that can be used for reproducible experiments, e.g. Reuters Corpora [47], MS MARCO [48], and TREC [49] among others, open cross-lingual data sets are all derived from Wikipedia. These include:

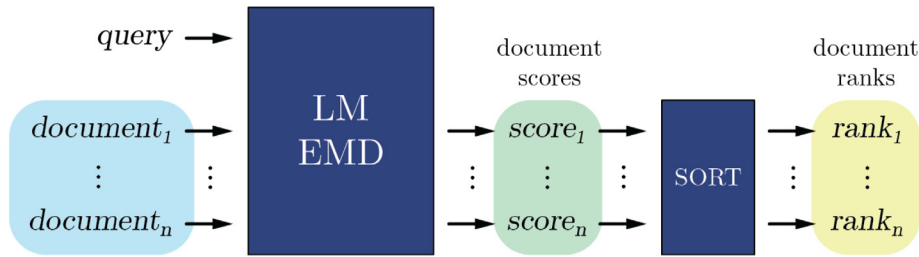


Fig. 2. The document ranking steps. The document's rank is assigned based on their EMD score given by the LM-EMD model.

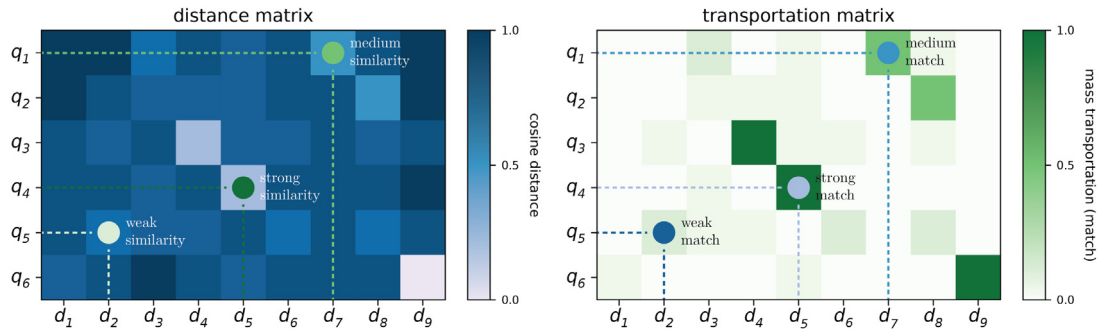


Fig. 3. Interpreting the score of a single document for a given query using the distance (left) and transportation (right) matrix. The colors in the matrices represent the intensity of the token distance (left) and their match (right), respectively.

- WikiCLIR [50]. A large-scale data set consisting of German queries and English Wikipedia articles as documents.
- CLIRMatrix [51]. A massive collection of bilingual and multilingual data sets. They collected 49M unique queries and 34B (query, document, label) triplets.
- WikiMatrix [52]. The data set contains 135M parallel sentences from 1.6k different language pairs in 85 languages.

To evaluate our model, we employ the Large-Scale CLIR Dataset [53]. This large-scale data set is derived from Wikipedia and consists of more than 2.8 million English queries and relevant documents from 25 selected languages and has been used to evaluate CLIR models. One could argue it is not suitable for evaluating multilingual language models since a majority of them are trained using Wikipedia. Nevertheless, we chose it to achieve reproducible results and ensure a healthy data set size.

Each row in the data set contains the following values:

- *Query*. The English query used for finding relevant documents. The query is the first sentence of the English article with the article's title removed.
- *Document*. The text of the document in the foreign language. The document is limited to the first 200 words of the article.
- *Relevance*. The relevance label between the query and the document. The possible values are 2 (relevant), 1 (somewhat relevant) and 0 (irrelevant). The “somewhat relevant” labels are not available in some language pair data sets.

While the training set is structured such that for each query there are 4 irrelevant document and 1 relevant document, the test set contains 39 irrelevant and 1 relevant document per query. In the experiments we use five languages pair data sets of varying degrees of similarity: English (EN) → German (DE), French (FR), Tagalog (TL), Japanese (JA), and Swahili (SW), respectively. The languages were selected to evaluate the models on: the same language family (EN, DE), different language families (FR, TL, SW, JA), low-resource languages (TL, SW), and a language that uses non-latin script (JA). The data sets are available at [54].

Table 1

The language pair set statistics. The ratio represents the number of relevant and irrelevant documents per query. While the German, French, and Japanese are larger, the Tagalog and Swahili language pairs contain significantly less queries and documents which is associated with their respective Wikipedia size.

Lang. pair	Training set			Test set		
	# queries	# docs	Ratio	# queries	# docs	Ratio
EN → DE	323,167	1,615,835	1/4	42,021	1,680,840	1/39
EN → FR	379,811	1,899,055	1/4	54,196	2,167,840	1/39
EN → JA	148,526	742,630	1/4	21,149	845,960	1/39
EN → TL	16,631	83,155	1/4	2359	94,360	1/39
EN → SW	7929	39,645	1/4	1160	46,400	1/39

5.2. Data preparation

For each data set, we retrieve all queries, documents, and relevance labels, and remove any trailing white-spaces from the queries and documents. Afterwards, the relevance labels are scaled to 1 (relevant) and 0 (irrelevant). The “somewhat relevant” labels are not present in these data sets. Table 1 shows the statistics of the selected language pairs.

5.3. Evaluation metrics

Following previous work [53], we use the P@1 (precision at 1) and MAP (mean average precision) metrics to measure the performance of the models.

The P@1 measures if a relevant document is found at rank 1. It is a special case of the P@k (precision at k) which corresponds to the number of relevant documents among the retrieved ones:

$$P@k = \frac{|\{\text{retrieved documents}\} \cap \{\text{relevant documents}\}|}{|\{\text{retrieved documents}\}|} \in [0, 1].$$

The MAP score is the mean of the average precision scores for each query $q \in Q$. The average precision score is defined as

$$\text{AveP}(q) = \frac{1}{R} \sum_{k=1}^n P@k \cdot \text{rel}@k,$$

where R is the total number of relevant documents, n is the number of retrieved documents, and $\text{rel}@k$ signifies if the document at rank k is relevant or not:

$$\text{rel}@k = \begin{cases} 1, & \text{the document at rank } k \text{ is relevant,} \\ 0, & \text{the document at rank } k \text{ is irrelevant.} \end{cases}$$

The mean average precision is then given by the equation

$$\text{MAP} = \frac{\sum_{q \in Q} \text{AveP}(q)}{|Q|} \in [0, 1].$$

Models with higher MAP scores assign higher ranks to relevant documents.

5.4. Implementation details

In this section, we present the details of how is the LM-EMD model developed. It is implemented using PyTorch [55] and its source code is available on GitHub [56].

Language model. The model uses the pre-trained multilingual BERT, more precisely the bert-base-multilingual-cased model,⁵ available via the HuggingFace's transformer library [57]. The language model consists of 110M parameters and was pre-trained on the top 104 languages with the largest Wikipedia using the masked language modeling objective. The multilingual BERT uses the WordPiece tokenizer [58], which adds special word boundary symbols such that the original word sequence can be retrieved from the wordpieces sequence without ambiguity.

Similarity measure. The Earth Mover's Distance is calculated using the Sinkhorn-Knopp algorithm [44], an iterative algorithm for solving regularized EMD. The algorithm requires two hyper-parameters: (1) the number of iterations nit , and (2) the regularization factor γ . The regularization factor is used to regulate the size of the regularization term and thus the precision of the final EMD optimization solution. Smaller factors provide more precise solutions, but they also introduces a higher risk of under- or over-flowing gradient values.

In order to provide the Sinkhorn-Knopp algorithm enough iterations to converge to a solution, we set $\text{nit} = 500$. To analyze the impact of the regularization factor on the model's performance, we train three models using $\gamma = \{0.1, 1, 10\}$, respectively. These values are frequently used with the algorithm in other related works [59].

Model fine-tuning. The LM-EMD model parameters are updated using the AdamW optimizer [60] with the initial learning rate $= 10^{-6}$, epsilon $= 10^{-6}$, and weight decay $= 0.01$. These hyper-parameters were selected using grid search and monitoring the loss values of the first 1000 examples from the EN $\rightarrow$ DE training data set.

The model fine-tuning is performed on the selected language pair training sets for 1 epoch. We employ gradient accumulation and update the model after every 16th mini-batch, where a mini-batch consists of the query and its 40 relevant and irrelevant documents. We use one of the two loss functions: (1) cross-entropy, and (2) pairwise ranking loss function, defined as

$$L(q, d^+, d^-) = \max \{0, S(q, d^-) - S(q, d^+)\},$$

where S is the scoring function (i.e. the EMD), q is the query, and d^+, d^- are the relevant and irrelevant documents, respectively. The model's fine-tuning and evaluation was performed on a Tesla V100 PCIe 32 GB graphic card.

5.5. Comparing models

In this section, we present the comparing models used to compare with the proposed LM-EMD. The models are selected to have an attribute similar to the proposed model, which is the possibility of using document representations generated ahead of time. Models that require both the query and document texts at inference time for constructing the input in the form of [CLS] query [SEP] document [SEP], such as those presented in some of the related work [27–29], causes massive computational overhead on larger data sets and are thus not considered in the evaluation.

NRM. The Neural Ranking Model [53] uses word embeddings and convolutional layers to generate the query and document vector representations. It was evaluated separately on the high-resource and low-resource language pairs of the Large-Scale CLIR Dataset. Since the model is already evaluated on the language pairs we are using in the experiment, we only report their best performances from the paper.

WE. These models use aligned word embeddings [19] to create the document vector representation. Following previous work [12,21], each query and document is represented as a weighted sum of its corresponding word embeddings:

$$\vec{v}_d = \sum_{t \in T_d} w_t \vec{x}_t,$$

where T_d is the set of the unique terms in document d , and w_t and $\vec{x}_t$ are the weight and word embedding associated with the term t , respectively. The similarity of two vectors is then measured with the cosine distance, where the smaller distance signifies greater similarity. We compare three different weighted sums where the weight w_t equals to:

- (i) **TF:** The *term frequency* of the term t in document d : $\text{TF}(t, d) = f_{t,d}$, where $f_{t,d}$ is the number of times the term t occurs in document d .
- (ii) **IDF:** The *inverse document frequency* of the term t within the training set D : $\text{IDF}(t, D) = \log \frac{|D|}{1 + |\{d \in D : t \in d\}|}$.
- (iii) **TFIDF:** The product of the *term frequency* and *inverse document frequency* of term t in document d and training set D : $\text{TFIDF}(t, d, D) = \text{TF}(t, d) \cdot \text{IDF}(t, D)$.

The queries are represented by the TF-weighted sum vectors. The embeddings used by the models are open source [61].

WE-EMD. The model uses the same aligned word embeddings as the WE models and EMD to calculate the similarity between the query and documents [42]. The EMD is solved using the Entropic regularized optimal transport [62] available in the POT Python library [63].

BERT. The models follow a similar approach as in the related work [30], where the query and document representation vectors are generated separately using the same multilingual BERT. The inputs are formatted to [CLS] input [SEP]. We compare three different vector representations:

- (i) **CLS:** The vector associated to the special [CLS] token.
- (ii) **MAX:** The positional maximum values across the output token vectors.
- (iii) **MEAN:** The mean of the output token vectors.

The document relevance is measured using the cosine distance, where the smaller distance signifies greater similarity. The models are fine-tuned using the same hyper-parameters as described in Section 5.4. In addition, we train the models using both cross-entropy and pairwise ranking loss to analyze their impact on the models' performance.

⁵ <https://huggingface.co/bert-base-multilingual-cased>.

Table 2

Model performances (P@1 and MAP scores reported). The bold values and values with an asterisk represent the best and second best performances on the given language pair, respectively.

Model	EN → DE		EN → FR		EN → TL		EN → JA		EN → SW	
	P@1	MAP	P@1	MAP	P@1	MAP	P@1	MAP	P@1	MAP
NRM [53]	.710	.820	.760	.850	.570	.690	.730	.840	.600	.730
PSOSL [65]	–	–	.438	.841	.596	.772	–	–	.600	.776
WE-TF [12]	.737	.836	.718	.823	.590	.704	–	–	–	–
WE-IDF [21]	.744	.840	.718	.822	.594	.712	–	–	–	–
WE-TFIDF [12]	.764	.854	.754	.846	.597	.710	–	–	–	–
WE-EMD [42]	.867	.919	.828	.895	.606	.723	–	–	–	–
BERT-CLS [30]	.978	.987	.978	.987	.851	.912	.955	.973	.913	.947
BERT-MAX [30]	.941	.964	.948	.969	.798	.874*	.912	.946	.824	.886
BERT-MEAN [30]	.967	.980	.958	.976	.786	.874*	.941*	.965*	.835	.897
LM-EMD (our model)	.977*	.986*	.974*	.985*	.801*	.874*	.955	.973	.890*	.932*

PSOSL. This end-to-end robust framework combines the pre-trained Polyglot [64] embeddings with the smooth cosine similarity measure and the smooth ordinal search loss objective function [65]. The model was evaluated on the Large-Scale CLIR Dataset, specifically on two high-resource languages (French and Italian) and two low-resource languages (Swahili and Tagalog). Although the authors of the model added 40 additional irrelevant documents for each query, changing the amount of data used in the experiments, we report their best performances from the paper.

6. Results

Table 2 shows the experiment results of the best performing models on the language pair data sets. Because aligned word embeddings were not available for the JA and SW languages, we omit the corresponding WE model performances from the results. In addition, because the original paper did not evaluate the PSOSL model for the DE and JA languages, and due to the limited resource we had for performing the experiments, their associated performances are not included. The reported LM-EMD is trained with pairwise ranking loss and regularization factor $\gamma = 0.1$.

The BERT-CLS model has the best performance across all of the data sets, followed by the proposed LM-EMD which is on par with it on the high-resource languages. A significant decrease in performance of all models is seen on the low-resource languages, especially on EN → TL. This is accredited to the limited resources that the low-resource languages have for training. Overall, LM-EMD outperforms BERT-MAX and BERT-MEAN, both using all token embeddings to measure the document's relevance.

While the BERT-CLS, BERT-MAX, and BERT-MEAN models return a single value representing the document's relevance, the LM-EMD outputs values that can be used to identify why a document is relevant. This makes the LM-EMD model superior in the sense of informativeness; in addition to measuring the relevance, it is also able to identify the tokens that make the document relevant. How the LM-EMD model identifies these tokens is described in Section 6.1.

6.1. Interpretability

This section describes how the LM-EMD model scores can be interpreted. We show that the document's relevance is associated with the number and intensity of the matches between the query's and document's tokens. In addition, the relevance can be described graphically.

We prepared the query “Who was the first president of the United States?” and five documents in German: one contains the answer (George Washington), one is similar but incorrect

(Abraham Lincoln), and three irrelevant documents with various degrees of similarity. Using the LM-EMD model we calculate all document relevance scores with respect to the query. The scores are shown in Table 3.

The model correctly detected that George Washington is the most relevant document, followed by Abraham Lincoln. To identify which tokens contributed the most to their respective score, we follow the interpretation described in Section 4.3, and generate the graphs of the distance matrix and transportation matrix values for each document (similar to Fig. 3). The graphs are shown in Fig. 4.

Looking the George Washington graphs (first row), we find that almost all query tokens have a strong match with those in the document. Most are literal translations (e.g. “president”, “Präsident”), but some are matched due to their semantic relatedness, such as the query word “Who” and document words “George Washington”.

The Abraham Lincoln graphs (second row) show that only half of the query tokens have a strong match. This is due the smaller number of close tokens as seen in the distance matrix. Because of this, the document got a worse relevance score than the George Washington sentence.

In comparison, the third set of graphs belongs to the Christopher Columbus document, which is an irrelevant document. The graphs show the model finds a strong match only for the literal translation of “first”, and some conjunction words, such as “was”, “the”, and “of”. Surprisingly, it also finds a weak match between the words “president” and “Vizekönig” (en. viceroy), which are positions of power, and between “United States” and “Kolumbus”.⁶ This shows that the model is able to detect the slightest relations between the cross-lingual tokens. Nevertheless, the document has a small number of matches and thus has a smaller relevance score.

As seen in the example, the document's relevance is associated with the number of strong matches between the query's and document's tokens; the greater the number and the stronger the match between the tokens, the greater the relevance. Because of this, the LM-EMD model generates interpretable relevance scores which are supported by the graphic visualizations as shown in this section.

6.2. Parameter analysis

We compare multiple LM-EMD models using different regularization factor values used by the Sinkhorn-Knopp algorithm. To assess the impact of the loss function on the model's performance, we trained models using the cross-entropy (CE) and pairwise ranking (PR) loss, respectively. The comparison results are found in Table 4.

⁶ A quick search shows that there are cities in the United States that are named Columbus, after Christopher Columbus.

Table 3
The relevance score of the German documents for the query “Who was the first president of the United States?” A smaller score means a greater relevance.

Question: Who was the first president of the United States?	
Document	Score
George Washington war von 1789 bis 1797 der erste Präsident der Vereinigten Staaten von Amerika.	0.6332
Abraham Lincoln amtierte von 1861 bis 1865 als 16. Präsident der Vereinigten Staaten von Amerika.	0.6773
Christoph Kolumbus wurde der erste Vizekönig der las Indias genannten Gebiete.	0.7392
Augusta Ada King-Noel, Countess of Lovelace, allgemein als Ada Lovelace bekannt war eine britische Mathematikerin.	0.7871
Marie Skłodowska Curie war eine Physikerin und Chemikerin polnischer Herkunft, die in Frankreich lebte und wirkte.	0.7926

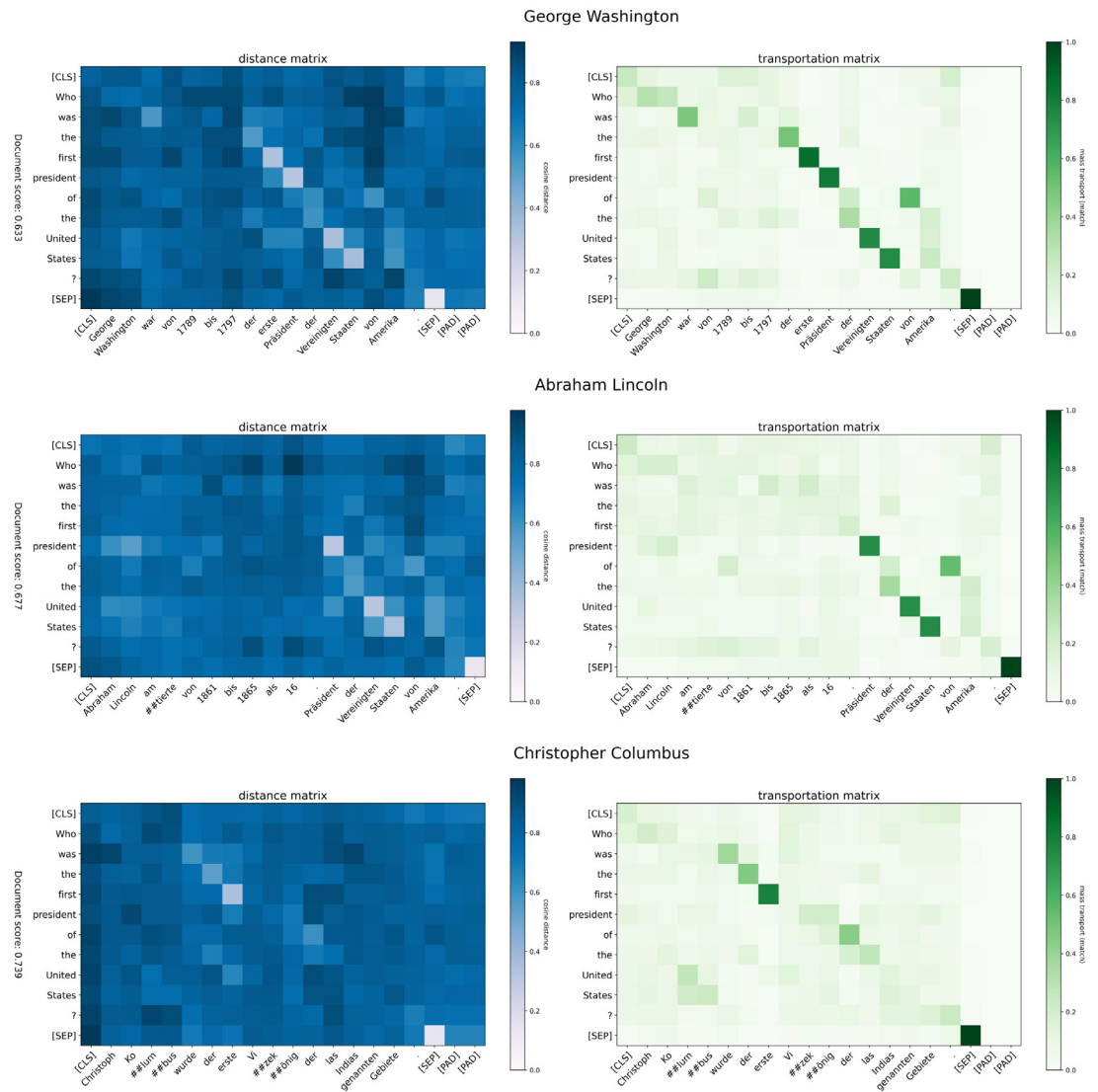


Fig. 4. The distance (left) and transportation (right) matrix of the first three documents (George Washington, Abraham Lincoln, and Christopher Columbus) from the example. The [PAD] tokens are ignored when calculating the EMD score.

The models using PR outperforms the models trained with CE, showing that PR matches better with the document retrieval evaluation metrics. The performance of all models trained with PR decrease with increasing regularization factor, except for the EN → TL language pair. The biggest change is seen on the JA and SW languages. While the decrease on JA can be because of the difference between the scripts used to represent the languages, the behavior of the SW can be accredited to its polysynthetic

language, e.g. having morphologically complex words with large number of affixes which EN does not have. In comparison, when trained with CE, the model's performance first decreases and then increases on low-resource languages, while on the high-resource ones the performance is best when using the largest regularization factor.

A similar analysis was done on BERT-based models giving similar results. Due to the limited resources we had for performing

Table 4

The performance comparison of the LM-EMD model trained with different regularization factor (γ) values, and using the pairwise ranking (PR) or cross-entropy (CE) loss function during training. The bold value highlights the biggest difference in performance (Δ) on the language pair.

Params	Loss	EN $\rightarrow$ DE		EN $\rightarrow$ FR		EN $\rightarrow$ TL		EN $\rightarrow$ JA		EN $\rightarrow$ SW	
		P@1	MAP	P@1	MAP	P@1	MAP	P@1	MAP	P@1	MAP
$\gamma = 0.1$	PR	.977	.986	.974	.985	.801	.874	.955	.973	.890	.932
	CE	.876	.927	.843	.909	.674	.793	.846	.908	.631	.754
	Δ	.101	.059	.131	.076	.127	.081	.109	.065	.259	.178
$\gamma = 1$	PR	.970	.982	.968	.981	.809	.883	.910	.946	.859	.913
	CE	.876	.927	.846	.910	.669	.790	.846	.907	.617	.747
	Δ	.094	.055	.122	.071	.140	.093	.064	.039	.242	.177
$\gamma = 10$	PR	.965	.979	.961	.978	.805	.881	.899	.941	.835	.899
	CE	.878	.928	.846	.911	.671	.792	.848	.909	.628	.753
	Δ	.087	.051	.115	.067	.134	.089	.051	.032	.207	.146

Table 5

The comparison of BERT-{CLS, MAX, MEAN} performances using the pairwise ranking (PR) loss and cross-entropy (CE) functions during training, with the difference in performance (Δ).

Model	Loss	EN $\rightarrow$ DE		EN $\rightarrow$ FR		EN $\rightarrow$ TL	
		P@1	MAP	P@1	MAP	P@1	MAP
BERT-CLS	PR	.978	.987	.978	.987	.851	.912
	CE	.893	.937	.855	.916	.729	.838
	Δ	.085	.050	.123	.071	.122	.074
BERT-MAX	PR	.941	.964	.948	.969	.798	.874
	CE	.861	.918	.824	.898	.676	.802
	Δ	.080	.046	.124	.071	.122	.072
BERT-MEAN	PR	.967	.980	.958	.976	.786	.874
	CE	.861	.918	.824	.898	.676	.802
	Δ	.106	.062	.134	.078	.110	.072

the experiments, we only report the results on EN $\rightarrow$ DE, FR, and TL language pairs. The analysis results are shown in Table 5.

7. Discussion

In this section, we discuss the experiment's results. We highlight the model's strong and weak points, as well as its theoretical and practical implications.

Model performance. The proposed LM-EMD model shows comparable performance with the best performing comparing model BERT-CLS on the high-resource languages with an average 96.8% P@1 and 98.1% MAP score. However, LM-EMD's performance is significantly lower than of BERT-CLS on low-resource languages, with an average 84.6% P@1 and 90.3% MAP compared to the average 88.2% P@1 and 93.0% MAP achieved by BERT-CLS.

When comparing LM-EMD to BERT-MAX and BERT-MEAN, both using all of the token embeddings to measure the document's relevance to the given query, the LM-EMD outperforms the other two models across all language pairs. This shows that EMD is a superior measure as it takes all of the token distances into account when assigning the relevance score. This is also reflected when comparing WE-based methods where WE-EMD has the best performance scores.

Interpretability. Unlike the cosine distance, EMD can be used to generate interpretable relevance scores. The EMD solution outputs the transportation matrix which indicates which query-document token pairs are semantically similar and thus improve the relevance score. The similarity it detects ranges from distant connections to synonyms and literal translations. However, EMD is more computationally expensive than cosine distance as it needs to run the Sinkhorn-Knopp algorithm to get the final score.

It is also prone to introducing higher risk of under- or over-flowing the gradient values when the regularization factor is too small. Because of this, the LM-EMD might be more appropriate to be used "per request", e.g. when the user would request an explanation on why a document is relevant to their query.

Loss function. The analysis of the impact the loss function has on the model's performance shows that models trained with pairwise ranking outperforms those with cross-entropy. Pairwise ranking is more appropriate for document retrieval models because it assesses if the order of the retrieved documents is correct, unlike cross-entropy which measures if the model is able to identify the document as relevant. While the Large-Scale CLIR Dataset has only one relevant document per query, other data sets might have multiple relevant documents, making pairwise ranking the more appropriate function.

8. Conclusion and future work

We propose a new LM-EMD model used for cross-lingual document retrieval. It generates outputs that both measure the document's relevancy with high precision and provide insight into why the document is relevant. It is compared to other word embedding and multilingual language model approaches showing it has similar performance as the best performing compared model on high-resource languages. We provide the explanation of how to interpret the model's outputs and show its capabilities on an example with an English query and German documents. An analysis of the impact an objective loss function has on training and fine-tuning CLIR models has shown that there is a significant increase in performance when using the pairwise ranking instead of the cross-entropy loss function. The code for reproducing the experiments is open source [56].

In the future, we intend to make the model smaller by using model distillation and pruning, which will potentially speed up the calculation of the relevance scores. Moreover we plan to integrate various text features, such as part-of-speech tags, into EMD calculation, replace EMD with p -Wasserstein distance where $p \geq 2$, and run the experiments using other multilingual language models. We believe this will increase the understanding of the usability of the proposed approach and potentially find new directions of how to make explainable models for (cross-lingual) document retrieval.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] L. Wang, Z. Luo, C. Li, B. He, L. Sun, H. Yu, Y. Sun, An end-to-end pseudo relevance feedback framework for neural document retrieval, *Inf. Process. Manage.* 57 (2) (2020) 102182.
- [2] J.A. Nasir, I. Varlamis, S. Ishaq, A knowledge-based semantic framework for query expansion, *Inf. Process. Manage.* 56 (5) (2019) 1605–1617.
- [3] F. Ensan, F. Al-Obeidat, Relevance-based entity selection for ad hoc retrieval, *Inf. Process. Manage.* 56 (5) (2019) 1645–1666.
- [4] S. Jain, K.R. Seeja, R. Jindal, A fuzzy ontology framework in information retrieval using semantic query expansion, *Int. J. Inform. Manage. Data Insights* 1 (1) (2021) 100009.
- [5] S. Kuzi, A. Shtok, O. Kurland, Query expansion using word embeddings, in: *International Conference on Information and Knowledge Management*, Proceedings, Association for Computing Machinery, 2016, pp. 1929–1932.
- [6] H.K. Azad, A. Deepak, Query expansion techniques for information retrieval: A survey, *Inf. Process. Manage.* 56 (5) (2019) 1698–1735.
- [7] R. Gao, C. Shah, Toward creating a fairer ranking in search engine results, *Inf. Process. Manage.* 57 (1) (2020) 102138.
- [8] S. Marchesin, A. Purpura, G. Silvello, Focal elements of neural information retrieval models. An outlook through a reproducibility study, *Inf. Process. Manage.* 57 (6) (2020) 102109.
- [9] F. Ture, E. Boschee, Learning to translate: A query-specific combination approach for cross-lingual information retrieval, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2014, pp. 589–599.
- [10] T. Mikolov, K. Chen, G. Corrado, J. Dean, Distributed representations of words and phrases and their compositionality, in: *Proceedings of the 26th International Conference on Neural Information Processing Systems - Volume 2*, Curran Associates Inc., 2013, pp. 3111–3119.
- [11] J. Pennington, R. Socher, C. Manning, Glove: Global vectors for word representation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2014, pp. 1532–1543.
- [12] I. Vulić, M.F. Moens, Monolingual and cross-lingual information retrieval models based on (bilingual) word embeddings, in: *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*, in: *SIGIR '15*, Association for Computing Machinery, 2015, pp. 363–372.
- [13] P. Bojanowski, E. Grave, A. Joulin, T. Mikolov, Enriching word vectors with subword information, *Trans. Assoc. Comput. Linguist.* 5 (2017) 135–146.
- [14] C. Xing, D. Wang, C. Liu, Y. Lin, Normalized word embedding and orthogonal transform for bilingual word translation, in: *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Association for Computational Linguistics, 2015, pp. 1006–1011.
- [15] M. Artetxe, G. Labaka, E. Agirre, Learning principled bilingual mappings of word embeddings while preserving monolingual invariance, in: *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2016, pp. 2289–2294.
- [16] S.L. Smith, D.H.P. Turban, S. Hamblin, N.Y. Hammerla, Offline bilingual word vectors, orthogonal transformations and the inverted softmax, in: *Proceedings of the International Conference on Learning Representations*, 2017.
- [17] A. Conneau, G. Lample, M. Ranzato, L. Denoyer, H. Jégou, Word translation without parallel data, in: *International Conference on Learning Representations*, 2018.
- [18] G. Glavaš, M. Franco-Salvador, S.P. Ponzetto, P. Rosso, A resource-light method for cross-lingual semantic textual similarity, *Knowl.-Based Syst.* 143 (2018) 1–9.
- [19] A. Joulin, P. Bojanowski, T. Mikolov, H. Jegou, E. Grave, Loss in translation: Learning bilingual word mapping with a retrieval criterion, in: *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2018, pp. 2979–2984.
- [20] T. Brychcin, Linear transformations for cross-lingual semantic textual similarity, *Knowl.-Based Syst.* 187 (2020) 104819.
- [21] R. Litschko, G. Glava, S.P. Ponzetto, I. Vulić, Unsupervised cross-lingual information retrieval using monolingual data only, in: *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, in: *SIGIR '18*, 2018, pp. 1253–1256.
- [22] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of deep bidirectional transformers for language understanding, in: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1 (Long and Short Papers), Association for Computational Linguistics, 2019, pp. 4171–4186.
- [23] W. Yang, H. Zhang, J. Lin, Simple applications of BERT for ad hoc document retrieval, 2019, arXiv, abs/1903.10972.
- [24] S. MacAvaney, A. Yates, A. Cohan, N. Goharian, CEDR: Contextualized embeddings for document ranking, in: *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, Association for Computing Machinery, 2019, pp. 1101–1104.
- [25] R. Nogueira, K. Cho, Passage re-ranking with BERT, 2019, ArXiv, abs/1901.04085.
- [26] Z. Akkalyoncu Yilmaz, S. Wang, W. Yang, H. Zhang, J. Lin, Applying BERT to document retrieval with birch, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP): System Demonstrations*, Association for Computational Linguistics, 2019, pp. 19–24.
- [27] W. Guo, X. Liu, S. Wang, H. Gao, A. Sankar, Z. Yang, Q. Guo, L. Zhang, B. Long, B.-C. Chen, D. Agarwal, DeText: A deep text ranking framework with BERT, in: *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*, Association for Computing Machinery, 2020, pp. 2509–2516.
- [28] S. Han, X. Wang, M. Bendersky, M. Najork, Learning-to-rank with BERT in TF-ranking, 2020, ArXiv, abs/2004.08476.
- [29] Y. Qiao, C. Xiong, Z. Liu, Z. Liu, Understanding the behaviors of BERT in ranking, 2019, ArXiv, abs/1904.07531.
- [30] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using siamese BERT-networks, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Association for Computational Linguistics, 2019, pp. 3982–3992.
- [31] T. Pires, E. Schlinger, D. Garrette, How multilingual is multilingual BERT? in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, 2019, pp. 4996–5001.
- [32] G. Lample, A. Conneau, Cross-lingual language model pretraining, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2019, pp. 7059–7069.
- [33] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, V. Stoyanov, Unsupervised Cross-lingual Representation Learning at Scale, in: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 8440–8451.
- [34] Z. Jiang, A. El-Jaroudi, W. Hartmann, D. Karakos, L. Zhao, Cross-lingual information retrieval with BERT, in: *Proceedings of the Workshop on Cross-Language Search and Summarization of Text and Speech (CLSSTS2020)*, European Language Resources Association, 2020, pp. 26–31.
- [35] C. Villani, *Optimal Transport: Old and New*, vol. 338, Springer, 2009.
- [36] C. Villani, *Topics in Optimal Transportation*, Vol. 58, American Mathematical Soc., 2021.
- [37] R. Saabni, Efficient word image retrieval using earth movers distance embedded to wavelets coefficients domain, in: *2013 12th International Conference on Document Analysis and Recognition*, 2013, pp. 314–318.
- [38] G. Yu, X. Xu, Z. Yan, H. Yu, Accelerating earth movers distance with instruction set extension for image retrieval, in: *2018 China Semiconductor Technology International Conference (CSTIC)*, 2018, pp. 1–3.
- [39] C. Zhang, Y. Cai, G. Lin, C. Shen, DeepEMD: Few-shot image classification with differentiable earth mover's distance and structured classifiers, in: *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 12200–12210.
- [40] S. Kumar, S. Chakrabarti, S. Roy, Earth Mover's Distance Pooling over Siamese LSTMs for Automatic Short Answer Grading, in: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI-17*, 2017, pp. 2046–2052.
- [41] J. Li, X. Liu, H. Zhao, R. Xu, M. Yang, Y. Jin, BERT-EMD: Many-to-many layer mapping for BERT compression with earth mover's distance, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2020, pp. 3009–3018.
- [42] L. Wu, I.E.H. Yen, K. Xu, F. Xu, A. Balakrishnan, P.Y. Chen, P. Ravikumar, M.J. Witbrock, Word mover's embedding: From Word2Vec to document embedding, in: *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2018, pp. 4524–4534.

- [43] L.V. Kantorovich, On the translocation of masses. (Dokl.) Acad. Sci. URSS 37 (3) (1942) 199–201.
- [44] G. Peyré, M. Cuturi, Computational optimal transport: With applications to data science, *Found.Trends[®] Mach. Learn.* 11 (5–6) (2019) 355–607.
- [45] R.M. Dudley, Distance of probability measures and random variables, *Ann. Math. Stat.* 39 (5) (1968) 1563–1572.
- [46] A.L. Gibbs, F.E. Su, On choosing and bounding probability metrics, 2002, [arXiv:math/0209021](https://arxiv.org/abs/math/0209021).
- [47] D.D. Lewis, Y. Yang, T.G. Rose, F. Li, RCV1: A new benchmark collection for text categorization research, *J. Mach. Learn. Res.* 5 (2004) 361–397.
- [48] P. Bajaj, D. Campos, N. Craswell, L. Deng, J. Gao, X. Liu, R. Majumder, A. McNamara, B. Mitra, T. Nguyen, M. Rosenberg, X. Song, A. Stoica, S. Tiwary, T. Wang, MS MARCO: A Human Generated MACHine Reading Comprehension Dataset, in: *Proceedings of the Workshop on Cognitive Computation: Integrating Neural and Symbolic Approaches 2016 Co-located with the 30th Annual Conference on Neural Information Processing Systems (NIPS)*, 2016.
- [49] F. Nanni, B. Mitra, M. Magnusson, L. Dietz, Benchmark for complex answer retrieval, in: *Proceedings of the ACM SIGIR International Conference on Theory of Information Retrieval*, Association for Computing Machinery, 2017, pp. 293–296.
- [50] S. Schamoni, F. Hieber, A. Sokolov, S. Riezler, Learning translational and knowledge-based similarities from relevance rankings for cross-language retrieval, in: *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics*, 2, Association for Computational Linguistics, 2014, pp. 488–494.
- [51] S. Sun, K. Duh, CLIRMatrix: A massively large collection of bilingual and multilingual datasets for cross-lingual information retrieval, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2020, pp. 4160–4170.
- [52] H. Schwenk, V. Chaudhary, S. Sun, H. Gong, F. Guzmán, WikiMatrix: Mining 135m parallel sentences in 1620 language pairs from wikipedia, in: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, Association for Computational Linguistics, 2021, pp. 1351–1361.
- [53] S. Sasaki, S. Sun, S. Schamoni, K. Duh, K. Inui, Cross-lingual learning-to-rank with shared representations, in: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Vol. 2, Association for Computational Linguistics, 2018, pp. 458–463.
- [54] K. Duh, Large-scale CLIR dataset, 2018, (accessed 29 October 2021), <https://www.cs.jhu.edu/%7Ekevinduh/a/wikiclr2018/>.
- [55] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, PyTorch: An imperative style, high-performance deep learning library, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2019.
- [56] E. Novak, The LM-EMD model, 2021, (accessed 29 October 2021), <https://github.com/ErikNovak/LM-EMD>.
- [57] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Brew, Transformers: State-of-the-art natural language processing, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Association for Computational Linguistics, 2020.
- [58] Y. Wu, M. Schuster, Z. Chen, Q.V. Le, M. Norouzi, W. Macherey, M. Krikun, Y. Cao, Q. Gao, K. Macherey, J. Klingner, A. Shah, M. Johnson, X. Liu, L. Kaiser, S. Gouws, Y. Kato, T. Kudo, H. Kazawa, K. Stevens, G. Kurian, N. Patil, W. Wang, C. Young, J.R. Smith, J. Riesa, A. Rudnick, O. Vinyals, G.S. Corrado, M. Hughes, J. Dean, Google's neural machine translation system: Bridging the gap between human and machine translation, 2016, *ArXiv*, [abs/1609.08144](https://arxiv.org/abs/1609.08144).
- [59] G. Peyré, M. Cuturi, Computational optimal transport, *Found. Trends Mach. Learn.* 11 (5–6) (2019) 355–607.
- [60] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: *International Conference on Learning Representations*, 2019.
- [61] Facebook Inc., Aligned word vectors fasttext, 2020, (accessed 29 October 2021), <https://fasttext.cc/docs/en/aligned-vectors.html>.
- [62] M. Cuturi, Sinkhorn distances: Lightspeed computation of optimal transportation distances, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2013, pp. 2292–2300.
- [63] R. Flamary, N. Courty, POT: Python optimal transport — POT python optimal transport 0.7.0 documentation, 2017, (accessed 29 October 2021), <https://pythonot.github.io/>.
- [64] R. Al-Rfou', B. Perozzi, S. Skiena, Polyglot: Distributed word representations for multilingual NLP, in: *Proceedings of the Seventeenth Conference on Computational Natural Language Learning*, Association for Computational Linguistics, Sofia, Bulgaria, 2013, pp. 183–192, URL <https://aclanthology.org/W13-3520>.
- [65] J. Liu, X. Zhang, D. Goldwasser, X. Wang, Cross-lingual document retrieval with smooth learning, in: *Proceedings of the 28th International Conference on Computational Linguistics*, International Committee on Computational Linguistics, Barcelona, Spain (Online), 2020, pp. 3616–3629, <http://dx.doi.org/10.18653/v1/2020.coling-main.323>, URL <https://aclanthology.org/2020.coling-main.323>.